Research Project: Franciscan Eco-Theology and Environmental Attitudes

A term paper submitted for the
RM in Management and Applied Microeconomics

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Abstract

Our project focuses on the effect of the exposure to two major Mendicant orders of the Middle Ages, the Franciscan and Dominican, between the 13th and 15th centuries on contemporary environmental attitudes. We find that that higher Franciscan exposure is associated with less pro-environmental attitudes, while higher Dominican exposure is associated with more pro-environmental attitudes contrary to our initial expectations. We also notice that the effect of the exposure to the two orders could be driven by the question on collective action, which is likely the underlying mechanism which is consistent with the contrasting theologies of the two orders regarding collectivity and individualism.

1 Introduction

Religious institutions, in particular the Church, played a pervasive role in every aspect of life in medieval Europe, shaping people's lives, values, and behavior throughout the Middle Ages (Lynch, 2014). One of the most famous religious figures of the Middle Ages is St. Francis of Assisi, who is the founder of the Franciscan Order. Being recognized by the Catholic Church as the patron saint of ecology, he is widely known for his deep affection towards nature and animals.

In this research, we study the Franciscan eco-theology and its effect on contemporary environmental attitudes. We have also controlled for the Dominican order, which is the other major Mendicant Order of the Middle Ages with comparably equal historical importance. Thus, our research question is the following: What is the effect of exposure to the Franciscan "Eco-Theology" from the 1200s to the 1500s on contemporary environmental attitudes among European populations (compared to the Dominican theology)?

To do so, we first construct our independent variables, the historical exposure measures for both Franciscan and Dominican Orders, defined as a fraction of a region's historical population living geographically close to a monastic house. We then construct our dependent variable, the environmental index, by applying Principal Component Analysis to the individual-level survey data on environmental attitudes from the European Values Study (EVS) 2017–2021 dataset. Using those variables and the set of controls, we empirically examine how historical exposure to the two Mendicant Orders affects contemporary environmental attitudes.

Given the historical background of St. Francis of Assisi, we have set the following initial hypothesis: we expected that higher historical exposure to the Franciscan Order is associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order is associated with less pro-environmental attitudes. However, our empirical results show the opposite. Contrary to our initial expectations, we find that higher historical exposure to the Franciscan Order is associated with less pro-environmental attitudes,while higher historical exposure to the Dominican Order is associated with more pro-environmental attitudes.

In this research, we have also proposed a possible mechanism that might have driven those results - the underlying differences in the approaches of both Catholic Orders on the collective action and individualism. The Franciscan theology of humility and acceptance of circumstances might have led to less pro-environmental attitudes, while the Dominican theology of persuasion and institutional coordination might have led to more pro-environmental attitudes.

2 Historical Context and Hypothesis

Both Franciscan and Dominican Orders shared substantial structural and institutional similarities. For example, both were Mendicant Orders, meaning that they relied on charity and begging for their livelihood and were more commited to preaching, teaching than owning landholding or monastic seclusion. Founded in the same century, both orders were operating as branches of the Catholic Church. Both orders were widespread across Europe, and had a significant influence on the medieval society. However, despite having substantial similarities, the two orders differ in their theological emphasis and approaches (Arrunada and Lopez-Manuel, 2024).

The Franciscan Order was founded in the early 13th century by St. Francis of Assisi (1181-1226), who was recognized by the Catholic Church as the patron saint of ecology, widely known for his profound affection towards nature and animals. This deep affection towards animals and nature can be seen in his "Canticles of the Creatures," where St. Francis refers to the sun and the moon as brother and sister. Importantly, Francis and his followers emphasized the close imitation of the life of Christ and the apostles through the ideals of poverty and humility. "He wanted to embrace real poverty, without any rationalisations or modifications. His band of followers would beg for their daily food, would take no care for tomorrow, and would own nothing – really nothing." (Lynch, 2014, p. 258). Thus, the Franciscan Order focuses on collective life, communal care, antimateralism, shame, compassion, simplicity, and humbleness (Arrunada and Lopez-Manuel, 2024).

In contrast, the Dominican Order, founded by St. Dominic (1170-1221) in the early 13th century, prioritized intellectual and educational mission of medieval Catholic Church. "For Dominic, a life of poverty and begging was a means to an end, whereas for Francis, it was an end in itself" (Lynch, 2014, p. 262). Having recognized that learning was important to train preachers, the Dominican Order had internal schools (Lynch, 2014). They also founded and staffed universities and produced extensive theological and practical writings (Le Goff, 1980; Boyle, 1981). The Dominicans emphasized the importance of critical thinking, self-examination, and moral development (Hinnebusch 1966; Lesnick 1989).

Because of the strong association of the Franciscan Order with nature and animals, we would naturally expect more-environmental attitudes in regions with stronger Franciscan exposure. Therefore, we initially hypothesized that higher historical exposure to the Franciscan Order would be associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order would be associated with less pro-environmental attitudes. However, our final results show the opposite.

3 Literature Review

A similar study was conducted by Arruñada and López-Manuel (2024), who examined the effects of the two major Mendicant Orders of the Middle Ages, the Dominican and the Franciscan, on the traits of the interpersonal exchange. Through their emphasis on guilt, shame, compassion, and analytical thinking, these orders shape outcomes at the cognitive, interpersonal, and institutional levels, which together constitute the "foundations" of impersonal exchange. They further argue that eco-theological beliefs of both Mendicant Orders "differ sharply in their approach to both spiritual and mundane life" and can foster distinct values and cultural norms (Arruñada and López-Manuel, 2024, p. 2). For example, at the interpersonal level, "Dominicans promoted guilt as a driver of prosocial behavior toward strangers", fostering a culture of individual accountability, independence, generalized fairness and trust, " whereas Franciscans encouraged compassion and shame, which strengthened bonds within the group but limited cooperation beyond it" (Arruñada and López-Manuel, 2024, p. 4). Consistent with these differences, the authors document contrasting effects of Dominican and Franciscan influence on impersonal exchange. Thus, their findings are consistent with our results regarding the contrasting effects of the two orders, as well as support our proposed mechanism that the underlying differences in theological beliefs might have driven the results.

Our study also relates to the broader literature on the role of medieval churches and persistance of cultural norms. The Church, as an institution, played a pervasive role in every aspect of life in medieval Europe, shaping people's lives, values, and behavior throughout the Middle Ages (Lynch, 2024). Once a religious institution becomes established, the values and norms it historically imposes tend to persist, as such cultural legacies "leave an imprint on values that endures" over time (Inglehart and Baker, 2000, p. 19). Although religious institutions have lost some of the formal allegiance of their followers in modern societies, Inglehart and Baker's findings suggest that religiously rooted value orientations continue to influence individuals, as these values become embedded in national cultures and are transmitted through educational institutions and mass media, allowing religiously rooted value orientations to persist at the individual level.

Another important strand of literature relates to the determinants of environmental attitudes. In much of the literature, it has been examined how cultural differences might influence environmental attitudes (Chwialkowska et al., 2020; Khan et al., 2024; Yang et al., 2024; Tam, 2025, as cited in Behler, 2025). Some literature also suggests that religiously grounded moral worldview of nature is associated with how people understand and relate to the environment (Cohen et al., 1976). Some theoretical models argue that pro-environmental attitudes may be rooted in self-interest and utility maximization. In particilar, a longer life expectancy may increase individuals' willingness to support environmental efforts to fight climate change (Mariani et al.,2010). Additionally, importantly for our research, many other studies with similar research interests commonly measure the environmental attitudes using the survey datasets (EVS) (Gelissen, 2007; Behler, 2025).

Taken together, these strands of the literature support and motivate our research, which examines the effect of historical exposure to Franciscan and Dominican monasteries on contemporary environmental attitudes.

4 Data and Methodology

4.1 Exposure Measure

In our reseach, we assume that people who are living close to the monasteries can get influenced or exposed to the theological beliefs of the monasteries. Thus, the exposure measure is defined as a fraction of a region's historical population living geographically close to a monastic house. To estimate historical exposure to the Franciscan and Dominican Orders, we employ the geodataset covering the years between 1216 to 1500 from the Mapping Past Societies (MAPS), which includes information on the geographic locations, the latitude and longitude, of the Dominican monasteries consisting of 870 observations and Franciscan monasteries consisting of 994 observations. The MAPS dataset reports foundation centuries for Dominican monasteries only, but not for Franciscan monasteries. So we assume that all monasteries in our dataset existed throughout the period of 1200-1500 since the majority of the monasteries were founded in the 13th century. However, this is also a major limitation of our approach, since it means that we are not able to capture the changes in the exposure over time, and we are only able to capture the overall exposure level. After we get the monastery location, we need to locate them within regions. To do so, we use the European Commision system for characterising geographical data, the Nomenclature of Territorial Units for Statistics (NUTS), as well as country shapefiles. NUTS-1 are for large macro regions, like German states, NUTS-2 are for more localized regions within countries. Since Germany has modified its NUTS administrative borders, we rely only on the larger NUTS-1 regions for Germany, and use NUTS-2 for other countries. We needed to delete monasteries in east Ukraine, Crimea, and Middle Eastern cities (Jerusalem/Acre), since those monasteries lie in locations that are not in or close to any NUTS region. There are some monasteries that are a little bit outside but still close to NUTS regions, so we coerce them to fit into the closest regions. As seen in Figure 1, the two Ukrainian monasteries were coerced to Eastern Romania since they are pretty close to it. Some Turkish monasteries appear in the dataset because the EU also collects data on parts of Turkey, although these observations are not included in the regression. They do not affect the results or procedure and we retain these observations for completeness. Similarly, we can also see monastary data in Greece as it appears in EU datasets. It should not affect the quantitative approach because there seems to be no entries for Greece in the EVS environmental survey data.

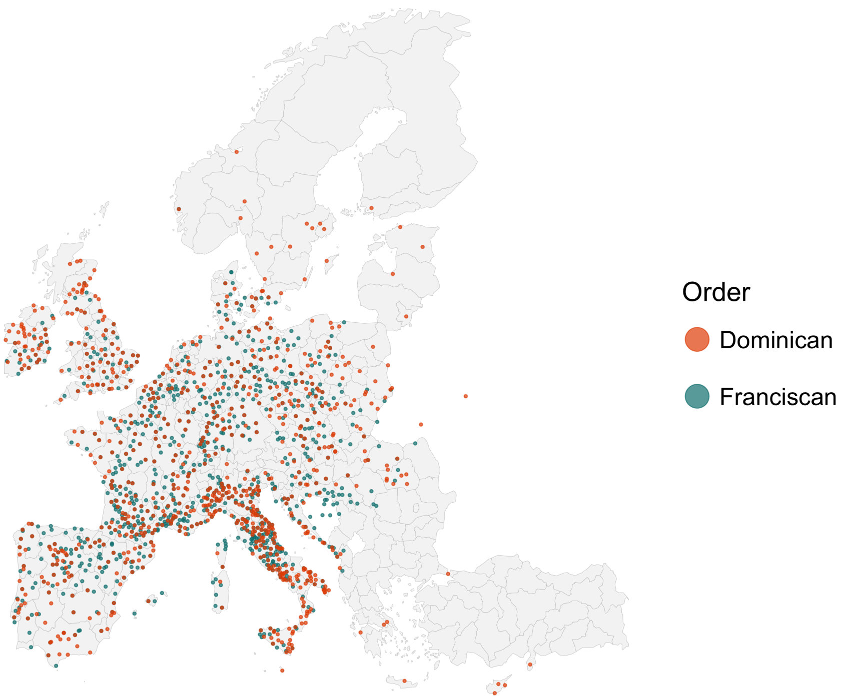
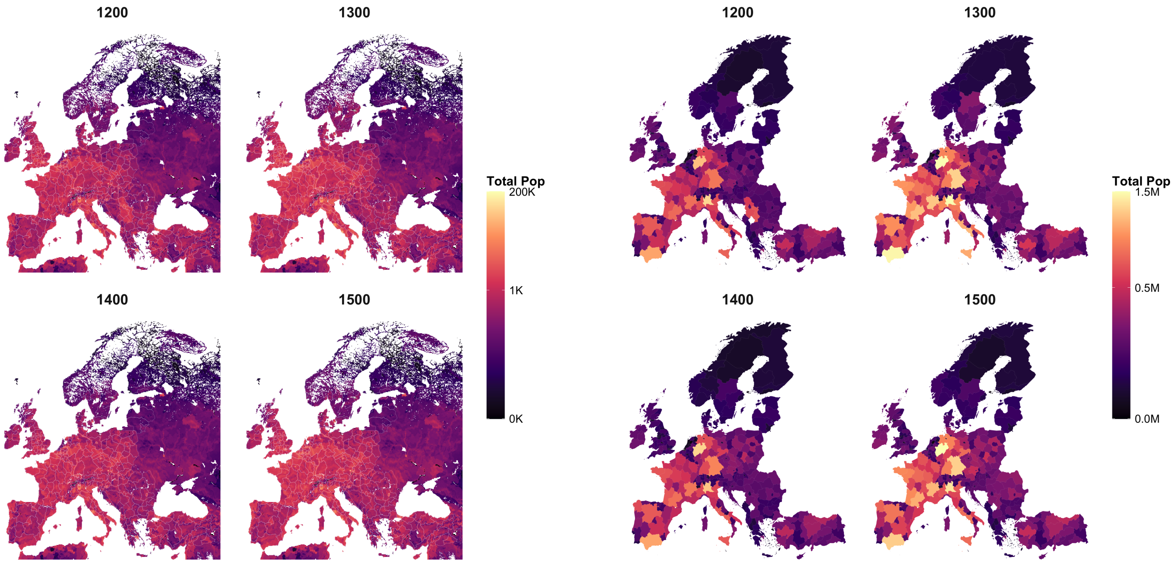


Figure 1: Locations of Franciscan and Dominican Monasteries

Once we have combined the monastery locations with NUTS and shapefiles, our next step is to observe historical population data.

HYDE 3.3 (History Database of the Global Environment) Utrecht University population data captures where people lived historically. It provides estimates of population counts and density on a regular latitude-longitude grid from the very late medieval time to the modern period. Grids are very small squares of 85 km^2, and we have population density estimates for each grid cell. In Figure 2(a), each cell represents estimated population counts for a historical year (1200–1500).



(a) Population densities (grid-level, log scale) (b) Regional populations (NUTS-2 level)

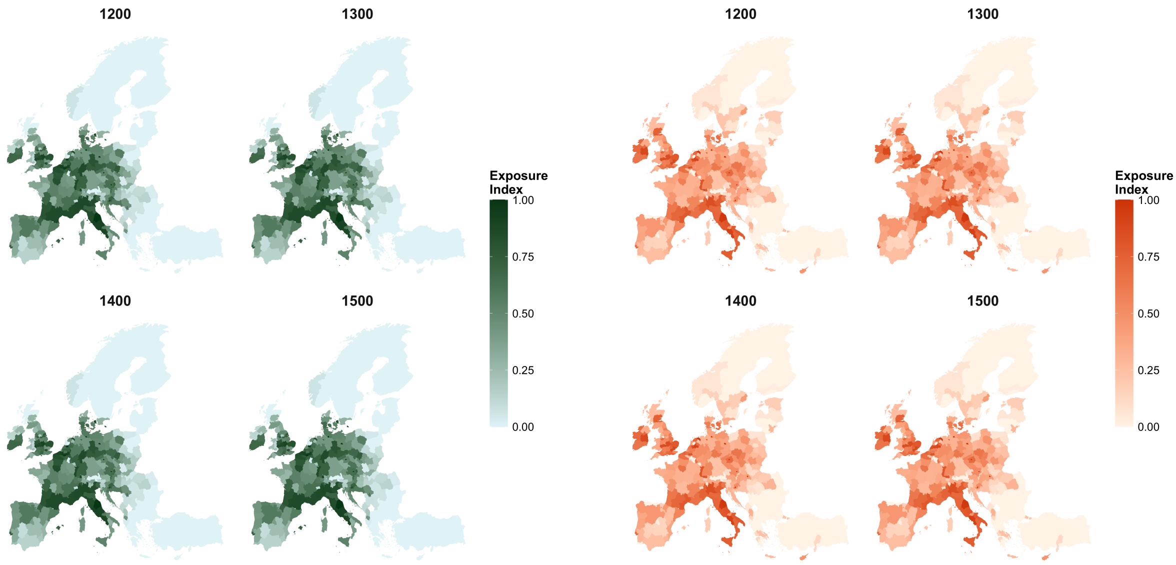
Figure 2: Population measures from HYDE 3.3

Similarly, we also consider the population densities not only in grid squares, but also at the NUTS regional level. In Figure 2(b), population data from HYDE 3.3 grids are overlaid with NUTS-2 boundaries where each grid cell is assigned to a modern NUTS region and the total population within a NUTS region is calculated as the sum of the population of all grid cells that fall within that region. If a grid cell is partially within a NUTS region, we assign the population of that grid cell to the NUTS region if more than half of the grid cell's area falls within the region. This way, we can estimate the population of each NUTS region based on the HYDE 3.3 grid data.

We draw a 25 km radius circle around the monastery, and all the little squares (grids) that are inside that circle are counted as the exposed population. All squares that are mostly (more than half) included in the circle are counted as fully exposed to the theology of that monastery. In our research, the exposure measure is defined as the sum of exposed population within a given region as a percentage of the total population for both Franciscan and Dominican monasteries:

$$\text{Exposure} = \frac{\text{Exposed Population}}{\text{Total Population}}$$

With all three data sources combined, we have Dominican and Franciscan exposure measures for Europe (Figures 3(a) and 3(b)).



(a) Franciscan exposure (b) Dominican exposure

Figure 3: Map of exposure to Mendicant Orders in Europe (1200–1500)
normalized index (0 = none, 1 = high exposure)

4.2 Environmental Attitudes Measure

To construct our main dependent variable, the environmental index, we use the European Values Study (EVS) dataset 2017–2021, which is individual-level survey data on values, beliefs, attitudes, and socioeconomic characteristics across European countries. The dataset covers over 30 European countries, of which 24 are relevant for our research when we exclude the countries that are not in NUTS-2. After data cleaning, the dataset includes 49233 observations and geographic identifiers for NUTS-1 regions for Germany and NUTS-2 for all other countries.

From the EVS dataset, we select six environmental variables in particular, which correspond to questions related to environmental attitudes. The questions are the following:

Question	**Dimension**
Choosing either “to protect the environment, even if slower economic growth and loss of jobs” or “to have economic growth and newly created jobs, even if the environment suffers”	Economy–environment trade-off
“I would give part of my income if I were certain that the money would be used to prevent pollution.”	Willingness to pay
“It is just too difficult for someone like me to do much about the environment.”	Self-efficacy
“There are more important things to do in life than protecting the environment.”	Priority of environmental protection
“There is no point in doing what I can for the environment unless others do the same.”	Collective action
“Many of the claims about environmental threats are exaggerated.”	Env. skepticism

The questions have categorical answers ("strongly agree", "agree", "neither agree nor disagree", "disagree", "strongly disagree", or choosing between the two statements). However, the questions differ in their directional interpretation. Thus, we need to standardize the questions by assigning a 0-1 scale for each answer to each question, such that the answers indicate the same environmental direction. In the standardized 0-1 scale, 1 indicates the least pro-environmental attitude, and 0 indicates the most pro-environmental attitude. See Figure 4 for visualization for environmental questions within the NUTS regions. The green regions indicate being more pro-environmental, while the yellow regions indicate less pro-environmental attitudes.

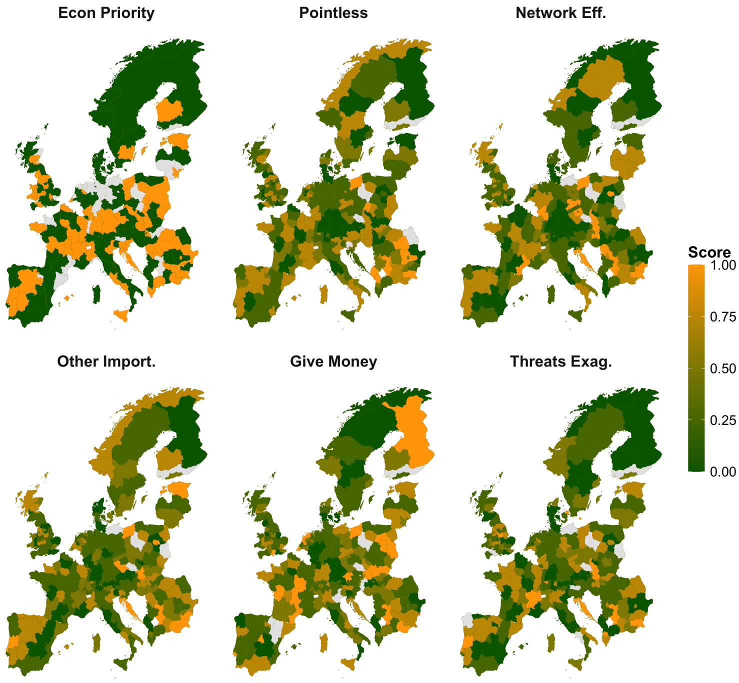


Figure 4: Distribution of the 6 environmental questions across Europe considering the average scores per region.

Since we have 6 different closely related questions for environmental attitudes, we want to reduce the dimensionality of the data and construct a single Environmental Attitudes Index. Thus, we run a Principal Component Analysis (PCA) to try to extract the main components of the questions, invisible or underlying mechanisms that are driving the answers. The first principal component explains 45% of the variance, and we take the first principal component as our Environmental Attitudes Index. We find that the components of Western European countries behave more green, while the Eastern European countries behave more yellowish as illustrated in Figure 5.

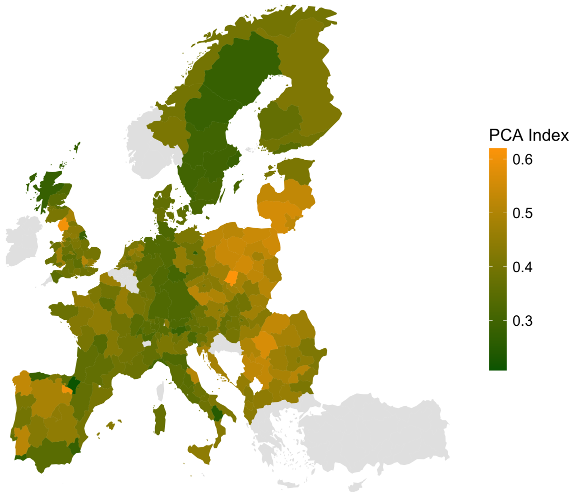


Figure 5: Environmental Index (PCA) across Europe
considering the average scores per region.

4.3 Regression Model

Once we have all the variables put together, we run our regression:

$$Y_{irc} = \alpha + \beta_1 Franc_r + \beta_2 Dom_r + X_i' \gamma + \delta_c + \varepsilon_{irc} \tag{1}$$

where Y_{irc} denotes the environmental attitude index for individual i in region r of country c (scaled 0-1), $Franc_r$ and Dom_r represent population-weighted exposure to Franciscan and Dominican Orders in region r measured in 1500, and X_i' is a vector of control variables, δ_c denotes country fixed effects, and ε_{irc} is the error term.

The control variables were also taken from the EVS data, the same dataset as the environmental question variables. The vector of individual-level control variables, X_i', consists of gender, age, education, household income (PPP-adjusted), socio-economic status (ISEI), town size, political orientation (left–right scale), and religious denomination with dummy variables for Catholic and Protestant. We cluster standard errors at the NUTS level.

5 Results

5.1 Regressions and Robustness Checks

We started by running the first four regressions shown in Table 1 and found the following results. Model 1 is the simplest form of the regression, where we regressed the environmental index on the Franciscan exposure only. And we see that the coefficient β_1 is negative and statistically significant. Because the environmental index ranges from 0 to 1, where lower values indicate greater pro-environmental behavior and higher values indicate less pro-environmental behavior, a negative coefficient indicates that the variable makes people more pro-environmental. β_1 in Model 1 was in accordance with our initial hypothesis that higher Franciscan exposure would make people more pro-environmental.

In Model 2, adding the Dominican exposure decreases β_1. When we add country fixed effects in Model 3, β_1 changes its sign and loses its significance. In Model 4, when we finally add all the individual-level controls on top of the Dominican exposure and country fixed effects, β_1 becomes fully positive and statistically significant, which means that regions that had maximum Franciscan exposure had a 3.5 percentage points lower environmental index. This is quite an interesting result for us. We could also see that many of the control variables have a significant effect. For example, females are more pro-environmental. However, the coefficient of the Dominican exposure, β_2, is not statistically significant in any of the models we have analysed. Even though we cannot say much about it, β_2 is always negative, indicating that generally the Dominican exposure is correlated with a more pro-environmental attitude.

Table 1: Regression Results (SE clustered by region)

	Model 1 Franciscans	Model 2 Fran vs Dom	Model 3 Country FE	Model 4 Full + FE
Franciscan Exposure	-0.095^{***} (0.016)	-0.058^{*} (0.027)	0.004 (0.014)	0.035^{*} (0.015)
Dominican Exposure	–	-0.050 (0.027)	-0.013 (0.014)	-0.016 (0.014)
Female	–	–	–	-0.028^{***} (0.003)
Age	–	–	–	0.000^{**} (0.000)
Education	–	–	–	-0.000^{***} (0.000)
Income (PPP)	–	–	–	-0.002 (0.001)
Right-wing	–	–	–	0.007^{***} (0.001)
Town Size	–	–	–	-0.005^{**} (0.001)
Socio-Economic Index	–	–	–	-0.001^{***} (0.000)
Catholic	–	–	–	0.004 (0.004)
Protestant	–	–	–	-0.001 (0.005)
Observations	49,233	49,233	49,233	24,379
R^2	0.022	0.024	0.099	0.149
Adj. R^2	0.022	0.024	0.099	0.148

Notes: Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

All the β_1 and β_2 coefficients from these regressions are plotted in Figure 6 (see Appendix). β_1 has a steady increase going from a negative to a positive sign as we move from Model 1 to Model 4, and finally becomes positive and statistically significant. β_2 is never significant; however, it also decreases once we add controls.

In Table 2 (see Appendix), we run regressions with each of the environmental questions as dependent variables separately to see whether there is one variable within the PCA index driving all of the results. We run the regression in Model 4 six times with one of the variables instead of the index. The values of β_1 and β_2 did not change that much across these six regressions compared to Model 4 in Table 1. However, there aren't any significant results for them except the variable indicating the network and collectivity effect, stating, "There is no point in doing what I can for the environment unless others do the same." The Franciscan and Dominican exposure coefficients produce significant results since they have contrasting theologies regarding this variable, where the Franciscan Order focused more on collectivity and the Dominican Order emphasized an individualistic view.

In Figure 7 (see Appendix), we run the main regression in Model 4 twenty-four times, each time excluding one country at a time to see whether there is one country driving the results. The plot on top shows the Dominican exposure coefficient, β_2, estimates across these regressions, and the bottom plot shows the Franciscan exposure coefficient, β_1, estimates. We see that there isn't much difference across countries. Only Switzerland and Italy have a slightly stronger effect on the results, but even excluding these countries, the results remain quite similar to those in Table 1. So the effects are homogeneous across Europe.

In Table 3 (see Appendix), we run the main regression again, this time excluding each control variable at a time for the sensitivity analysis. We see that the estimate of β_1 remains around 0.035 and is statistically significant across most of the regressions, except for one regression in which we exclude the town size control variable, which accounts for urbanization. This is an important result because the Franciscans used to locate themselves within cities and urban areas to reach more people. So, not controlling for the town size would bias the coefficient of Franciscan exposure. This makes the town size a very important control, but apart from it, we could drop any other control and find very similar results to our full model.

In Figure 8 (see Appendix), we see the plotted regression coefficient estimates found in the sensitivity analysis in Table 3 by leaving out one control in each regression, the top plot showing estimates for β_2 and the bottom one showing estimates of β_1. We can clearly see that only the town size shifts the Franciscan exposure coefficient estimate. The Dominican exposure coefficient estimate also changes slightly when we remove income, political orientation, and town size, but none of the controls change the general direction of the results.

As the last robustness check in Table 4 (see Appendix), we run subsample regressions with the full model, including all controls, but keeping only Catholic individuals in the first subsample. The Catholicism of individuals is determined from the same EVS data by asking the question "Which religious denomination do you belong to?" and taking into account only those who answered with the option "Roman Catholic". Considering only Catholics, the sample size drops significantly to 5899, so we don't have any significant results and can't say much about them. In the second subsample, we exclude the observations from Germany to only keep NUTS 2 regions to see whether it was the lack of linearity in Germany driving the results, since we consider NUTS 1 regions for Germany. But we see that the results stay pretty much the same as in the main model.

5.2 Possible Mechanism

Since we found contrasting results to what we were expecting to find, we didn't want to leave it like this and want to propose a possible mechanism that might be driving these results, which could be used for further hypotheses and research. We have already ruled out all misleading mechanisms that could have come from the control variables, such as education, income, and political views, and from country fixed effects. For example, one could have claimed that Dominican exposure also meant a higher education level since they promoted teaching, which could have led to more pro-environmental attitudes. However, the mechanism should be a deep, slow-moving, cultural or institutional effect.

We propose that the mechanism could lie in the opposite beliefs about taking collective action within the Franciscan versus Dominican theologies. The Franciscan theology is the theology of humility and acceptance of circumstances, poverty, and scarcity for the sake of imitating Christ. They value moral purity over institutional action. The lack of desire for collective intervention might have shown itself through having less pro-environmental attitudes, which could also be observed in our variables, beliefs about individual efforts being pointless unless others do the same, and environmental threats being exaggerated. The Franciscans are not likely anti-nature but rather anti-intervention. The Dominican theology, on the other hand, is the theology of persuasion and institutional coordination. Their willingness to take reasoned collective action and enforce norms could have translated into more modern pro-environmental attitudes. Their sense of obligation also possibly played a role in the context of the environment, and they might have had greater trust in institutions for collective solutions against environmental threats.

6 Conclusion

In this research, we analyzed whether the exposure to the two major Mendicant Orders of the Middle Ages, Franciscan and Dominican monasteries, has an effect on contemporary environmental attitudes. We constructed an exposure measure by combining the geo-dataset of monastery locations with historical population data, and we constructed an environmental attitudes index using the European Values Study survey data on individual environmental questions. We ran regressions controlling for various individual-level characteristics and country fixed effects. Through this, we have found that higher Franciscan exposure is associated with less pro-environmental attitudes, while the Dominican exposure is associated with more pro-environmental attitudes. More specifically, in our main regression model, β_1 is 0.035 and statistically significant, while β_2 is -0.016 and not statistically significant. Through various robustness checks, we have found that the results are quite robust and homogeneous across Europe and persist even when we exclude certain controls or countries.

We also proposed a possible mechanism that could be driving these results, which is the contrasting beliefs about collective action within the Franciscan and Dominican theologies. The Franciscan theology of humility and acceptance of circumstances might have led to less pro-environmental attitudes, while the Dominican theology of persuasion and institutional coordination might have led to more pro-environmental attitudes. The results might have been driven by underlying beliefs about whether the environmental threats are exaggerated, whether individual efforts are pointless unless others do the same, and whether individual efforts are important for the environment.

We can conclude our results by saying that long-run cultural and theological beliefs and monasteries can have a significant influence on contemporary attitudes toward the environment. The results of this research are quite interesting and surprising, but we cannot help but mention the limitations of our research and the need for further research.

7 Limitations and Further Research

The main limitation of our research is the Mapping Past Societies (MAPS) dataset, which contains foundation centuries only for Dominican monasteries, but not the specific dates, and it doesn't contain closing dates for any monasteries. Thus, we had to assume that all the monasteries were active throughout the period of 1200-1500, and so our exposure measure treats monastery presence as time-invariant.

The main extension that we propose is to find data on the foundation and closing dates of the monasteries to construct a more accurate exposure measure. We propose to use another existing dataset from Boranbay and Guerriero (2019) specifically for the Franciscan monasteries. They use a novel dataset which contains 2980 observations (while only 994 observations in MAPS) with specific foundation dates and monastery names only for Franciscan monasteries. But all of the closing dates are indicated as 2011, which means that there are still no closing dates available in this dataset. Still, this would allow us to construct a more accurate exposure measure for the Franciscan monasteries.

Another limitation is that our specification is purely historical and correlational. We cannot fully rule out the deep unobserved regional culture, so we cannot make any causal claims without a proper experiment. We should think about what the ideal experiment would look like. This would help us understand the data limitation and the choices that we are making. The ideal experiment would be to randomly drop a Franciscan monastery and then see what happens to these people who live there over time relative to everybody else. But for that we need the random assignment of monasteries. There is also a problem with the fact that there are no Franciscan monasteries in some NUTS regions in Northern Europe, and also no Dominican monasteries in some NUTS regions in Eastern Europe, so we cannot compare the effects of the two orders in those regions. An easy thing to do here would be to exclude the regions that do not have either of the monasteries.

Another major issue is that we rely on data from 800-500 years ago and assume that the influence of the monasteries has persisted until today on the local populations, but don't know whether the people in the EVS dataset who live there now are the ones whose ancestors were effected or exposed to the monasteries in those regions during the Middle Ages. So as an extension, we would have to think about migration flows. We could try to find data on the population migration and mobility across Europe to see how much of the population is actually exposed to the monastic influence. At least, it could be possible to account for the very important migration flows, for instance, in Germany, there was a huge influx from Eastern Europe after WWII.

One thing that we would also do if we had more time, is to run the regression specification that we have in Model 4 (main model) with the sample size that we have in Model 1, which is the full sample without any missing values for the control variables. We lose a lot of observations in Model 4 when we include controls because there are many unreported control variables. But then it becomes hard for us to judge whether it is the inclusion of the control variables that flips the coefficients, or whether it is the sample composition. Or we could also run Model 1 with the sample size of Model 4 including only observations with all control variables. It would be good to know how much of the change is driven by sample composition and how much of it is driven by the inclusion of controls.

Another thing that needs to be discussed is whether all of these controls are good controls. Some of them might be bad controls. For example, it might be that because of the Franciscan exposure people are less likely to be right-wing. Some of the effect of the exposure might be biased because of the control variables, and then we would call these bad controls. This is probably not the case for the controls age and gender, but it might be the case for all the other controls, such as education, income, political orientation, and town size. We could run the regression with only good controls and see how much of the effect is driven by the bad controls.

As other possible extensions, we could check for the choice of monastery locations. It is possible that monasteries were not randomly located across Europe. So we could construct insturumental variables for the monastery locations, such as the distance to the nearest city or trade route, use additional controls, or explicitly model the endogenous placement to improve the causal identification and remove the endogeneity problem. We could also construct a distance-weighted or decay-based exposure measure. Instead of assuming that everyone within a 25 km radius of the monastery is fully exposed to the theology of that monastery, we could assume that the influence decays with distance and people are less exposed the further they are. We would also like to test the effects of monasteries on other outcomes, such as climate policy support, trust in institutions, and collective action. By using other datasets, we could extend our research to other countries outside of Europe, such as Latin America, where the influence of the Catholic Church is also quite strong. For example, Waldinger (2017) uses a Mexican dataset to show the long-term effects of missionary orders, such as the Franciscans, Dominicans, and the Jesuits, on outcomes such as education and Catholicism.

In summary, due to our quite challenging cultural research and working with historical data, our paper so far has many issues and limitations, and it is important to acknowledge them. It is important to go over the choices that we make in this specification to try to think about what the ideal experiment is, even though oftentimes we do not have the data to do that experiment. As we think about where we make choices, we would think about where we deviate from the ideal experiment for many reasons such as not having ideal data. This is a good way of thinking about the problem and seeing our limitations. Because maybe the source of some awkward findings could be these choices. And only when we believe all of these choices and assumptions can we trust the results. So then the next step in further research could be to get better data and try to solve particular problems and limitations. We hope that our research can be a starting point for further research on the long-term effects of religious institutions on environmental attitudes and other outcomes.

8 References

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9 Appendix

Figure 6: Impact of Monastic Orders (Clustered SEs)
95% CI clustered by region. Crossing 0 = Not Significant

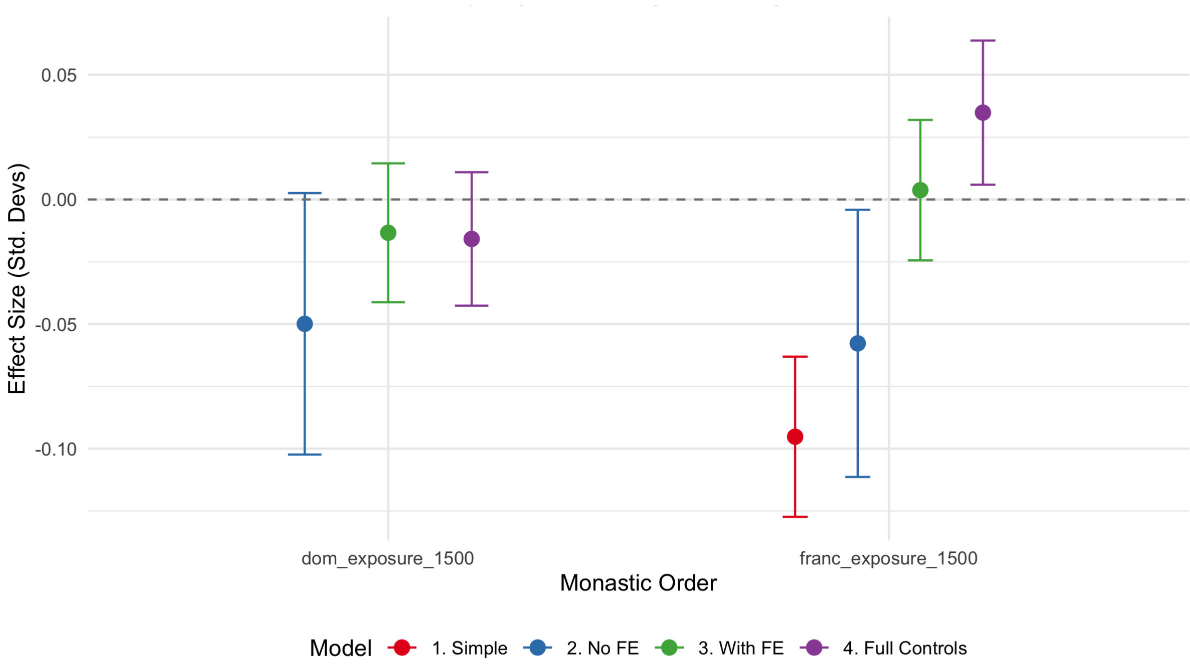

Table 2: Individual Environmental Questions (Clustered SEs)

	(1) Econ Priority	(2) Give Money	(3) Too difficult	(4) Other Import.	(5) Collective	(6) Threats Exag.
Franciscan Exp.	0.052	0.016	0.034	0.030	0.064**	0.037
	(0.038)	(0.021)	(0.021)	(0.019)	(0.021)	(0.021)
Dominican Exp.	-0.020	-0.026	-0.014	-0.018	-0.039*	-0.017
	(0.037)	(0.021)	(0.018)	(0.016)	(0.016)	(0.015)
Gender (Female)	-0.027***	-0.002	-0.023***	-0.040***	-0.028***	-0.047***
	(0.007)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)
Age	0.001**	0.001***	0.002***	0.000*	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Education	-0.000**	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Income (PPP)	-0.002	-0.004*	-0.011***	-0.004**	-0.010***	-0.004**
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
Socio-Econ. Ind.	-0.002***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Town Size	-0.003	-0.006**	-0.001	-0.004*	-0.009***	-0.012***
	(0.004)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Polit. Orient. (R)	0.011***	0.006***	0.003**	0.004***	0.005***	0.010***
	(0.003)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Catholic	0.007	0.005	0.011*	0.010	0.017*	0.007
	(0.009)	(0.007)	(0.006)	(0.005)	(0.007)	(0.007)
Protestant	-0.003	0.003	-0.007	-0.017**	-0.004	0.008
	(0.013)	(0.012)	(0.006)	(0.005)	(0.006)	(0.008)
Num. Obs.	25,427	27,212	27,344	27,370	27,343	26,823
R^2	0.095	0.121	0.140	0.141	0.125	0.121
Adj. R^2	0.094	0.120	0.139	0.139	0.124	0.120

Notes: Standard errors in parentheses. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 3: Sensitivity Analysis: Coefficients when Dropping Controls

	Full Model	Excl. Gender	Excl. Age	Excl. Education	Excl. Income	Excl. Politics	Excl. Town Size	Excl. Socio-Econ. Index	Excl. Religion
Franc. Exp.	0.035*	0.036*	0.035*	0.035*	0.032*	0.033*	0.012	0.039**	0.034*
Dom. Exp.	-0.016	-0.016	-0.015	-0.017	-0.008	-0.010	-0.010	-0.020	-0.016
Num. Obs.	24,379	24,379	24,434	24,438	26,795	33,560	28,220	27,286	24,379
R^2	0.149	0.144	0.149	0.147	0.143	0.135	0.141	0.148	0.149
Adj. R^2	0.148	0.143	0.147	0.145	0.142	0.134	0.139	0.147	0.148

Notes: $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 7: Does one country drive the results?
Coefficients when excluding each country one at a time (Clustered 95% CI)

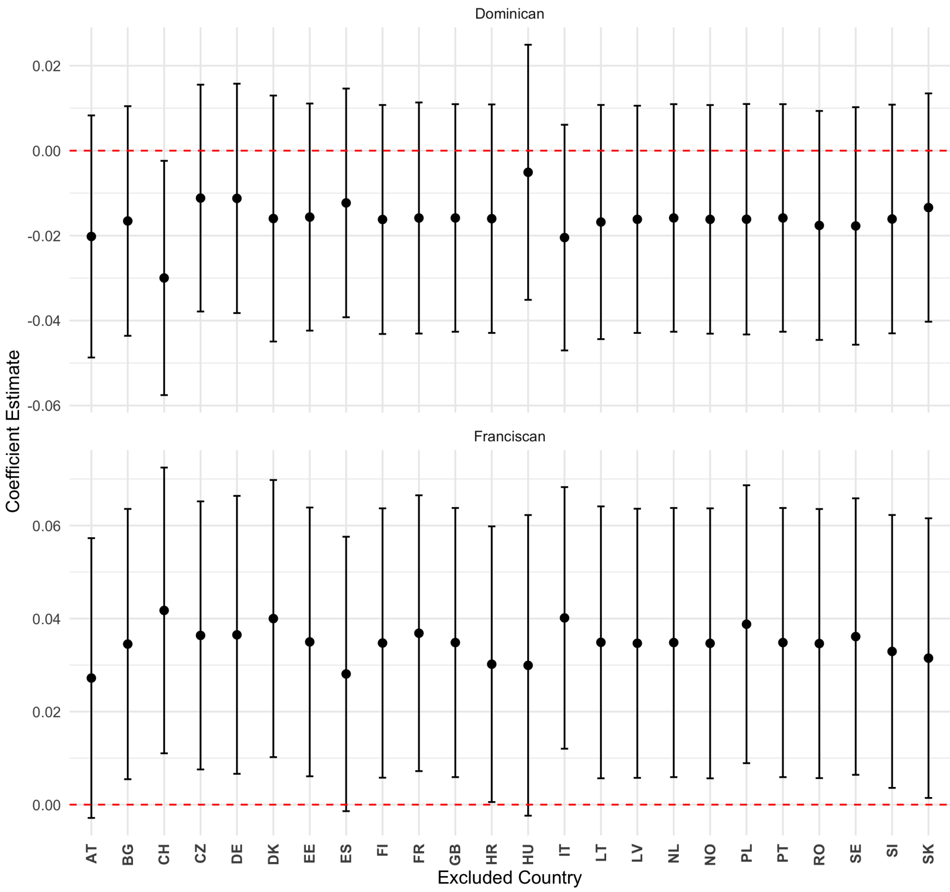

Table 4: Robustness Checks: Subsamples

	Main Model (All)	Catholics Only	Exclude Germany
Franciscan Exposure	0.035*	0.023	0.037*
	(0.015)	(0.021)	(0.015)
Dominican Exposure	-0.016	0.004	-0.011
	(0.014)	(0.018)	(0.014)
Num. Obs.	24,379	5,899	23,182
R^2	0.149	0.132	0.146
Adj. R^2	0.148	0.126	0.145

Notes: Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 8: Sensitivity Analysis: Leave-One-Control-Out
Does dropping a specific control variable change the results?

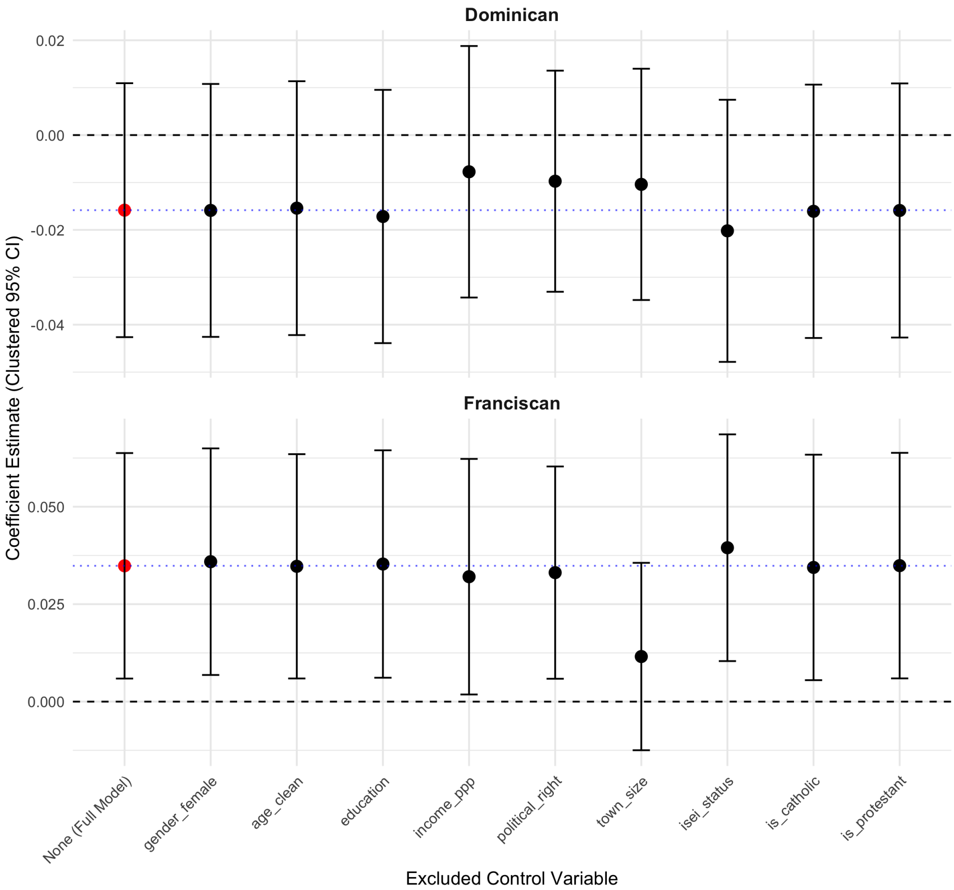

Table 5: Environmental Index: PC1 Loadings

	PC1 Loading
envir_econ_priority	0.855
envir_efforts_pointless	-0.287
envir_other_importances	0.498
envir_network_effect	0.200
envir_threats_exaggerated	0.510
envir_protection_money	0.085

	Value
SS Loadings (PC1)	1.368
Proportion of Variance Explained	0.228